1. (30 points) In this exercise, we consider two decks of cards: a "black" deck of 52 cards (containing the values $2, \ldots, 10$, J, Q, K, and A, for each of the four suits[1]), and a "red" deck of 32 cards (containing only $7, \ldots, 10$, J, Q, K, and A, of the four suits).

   (i) (6 points) We perform the following procedure: we toss a fair coin to select one of the two decks, and we then pick two cards simultaneously (i.e., at the same time), uniformly at random, from the selected deck. Compute the probability that the two cards are of the same value.

   (ii) (4 points) We observe that the two cards are of the same value: compute the probability that the black deck was selected.

   (iii) (4 points) We now merge and shuffle the two decks, to produce a deck of $52 + 32 = 84$ cards. We then pick two cards simultaneously, uniformly at random, from that large deck. Compute the probability that the two cards are of the same value.

   (iv) (4 points) We successively pick five cards from the 84-card deck, with replacement: each of them is chosen uniformly at random, and then put back into the deck, before picking the next card. Let X_B be the number of black cards that are picked, and X_R be the number of red cards. Determine the distribution of X_B, and the distribution of X_R.

   (v) (6 points) Let $X = X_B - X_R$. Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

   (vi) (6 points) We now pick successively $n = 7000$ cards (still with replacement), and we count the number X_{JQK} of cards which are either J, Q or K (i.e., whose value belongs to $\{J, Q, K\}$). Give an estimate of $\mathbb{P}(X_{JQK} \geq 2100)$ using the Central Limit Theorem, in terms of the cumulative distribution function Φ of a random variable with the standard normal distribution.

2 . (30 points) We consider a six-sided die, which is balanced (all integers between 1 and 6 are equally likely to appear). We roll it twice, and we assume that the outcomes of the two rolls are independent. Consider the events $E = \{\text{the two rolls produce different numbers}\}$, $F = \{\text{the second roll produces 5}\}$, and $G = \{\text{the sum of the two rolls is even}\}$.

   (i) (4 points) Compute $\mathbb{P}(E)$.

   (ii) (6 points) Are the two events E and F independent?

   (iii) (6 points) Are the two events E and G independent?

   (iv) (6 points) Compute $\mathbb{P}(G|E)$ and $\mathbb{P}(G|E^c)$.

   (v) (4 points) Are the three events E, F and G independent?

   (vi) (4 points) We now repeat the random experiment (of rolling twice the die) until we get the same number in the two rolls. We denote by T the number of times that the random experiment had to be performed (so that we had to roll the die $2 \cdot T$ times): what is the distribution of T? Give the probability mass function of T.

3 . (20 points) On an assembly line for laptops, keyboards are coming from four suppliers, that we call A, B, C, and D: 40% are coming from A, 25% from B, 20% from C, and 15% from D. It is estimated that among the keyboards supplied by A, B, C, and D, respectively 0.02%, 0.03%, 0.07%, and 0.06% are defective.

(i) (4 points) Compute the probability that a keyboard chosen at random is defective.

(ii) (6 points) We pick one keyboard at random, and it happens to be defective: compute the conditional probability that it was produced by each of the four suppliers.

(iii) (4 points) On a given day, $n = 10^4$ keyboards are used, and we denote by N the number of defective ones. We assume that the "states" (defective or not) of these keyboards are independent of each other. What is the distribution of N?

(iv) (6 points) Compute the expectation and the standard deviation of N.

4 . (30 points) We perform the following random experiment, with a biased coin which shows up "Heads" with probability p, and "Tails" with probability $1 - p$, for some $p \in [0, 1]$. We choose an integer N uniformly at random in $\{1, 2, 3, 4\}$ (i.e. all integers between 1 and 4 are equally likely). We then toss the coin N successive times, in an independent fashion, and we let X denote the number of times that we obtain "Heads".

(i) (4 points) What is the conditional distribution of X given $N = 2$?

(ii) (8 points) Give, in a table, the joint probability mass function of (N, X).

(iii) (4 points) Are N and X independent? Justify.

(iv) (4 points) Let p_X denote the probability mass function of X. Compute $p_X(2)$.

(v) (6 points) Find the conditional distribution of N given $X = 3$.

(vi) (4 points) We now consider the random variable $Z = \mathbb{1}_{X \text{ is even}}$. Among the following values for p, indicate those for which N and Z are independent: $p = 0$, $p = \frac{1}{4}$, $p = \frac{1}{2}$, $p = \frac{3}{4}$, $p = 1$. No explanation is required.

5 . (20 points) We consider a sample of n i.i.d. random variables $(X_1, \ldots, X_n)$, each uniformly distributed on $[-\theta, \theta]$, where $\theta \in (0, +\infty)$ is an unknown parameter.

(i) (6 points) Compute the expectation $\mathbb{E}[X_i]$ and $\mathrm{Var}(X_i)$, for each $1 \leq i \leq n$.

(ii) (6 points) Find an estimator $\widehat{\theta^2}$ of θ^2 using the method of moments.

(iii) (8 points) Compute the bias and the mean squared error of the estimator found in the previous question.

6 . (30 points) We consider two jointly continuous random variables X and Y, with joint probability density function

$$f_{(X,Y)}(x, y) = \frac{3}{2} xy \mathbb{1}_{x \geq 0} \mathbb{1}_{y \geq 0} \mathbb{1}_{x+y \leq 2}.$$

(i) (6 points) Compute $\mathbb{P}(X + Y \leq 1)$.

(ii) (4 points) Are X and Y independent? Explain why.

(iii) (6 points) Compute the marginal probability density function of X.

(iv) (6 points) Compute $\mathbb{E}[X]$.

(v) (8 points) Compute $\mathrm{Cov}(X, Y)$.

7 . (30 points) Let $\theta \in \mathbb{R}$ be given. We consider a random variable X with probability density function

$$f_X(x) = \frac{3}{(x-\theta+1)^4} \cdot \mathbb{1}_{[\theta,+\infty)}(x).$$

(i) (6 points) Compute $\mathbb{E}[X]$.

(ii) (8 points) Compute $\mathrm{Var}(X)$.

(iii) (6 points) We now consider a sample $(X_1, \ldots, X_n)$ of size $n \geq 1$: the random variables $(X_i)_{1 \leq i \leq n}$ are independent, and each is continuous with the same distribution as X, i.e. with probability density function f_X. Find an estimator of θ by the method of moments.

(iv) (4 points) Compute the bias of the estimator found in the previous question.

(v) (6 points) Determine the maximum-likelihood estimator $\hat{\theta}_{\mathrm{MLE}}$ of θ. Is it biased?

8 . (30 points) Let $T \in \mathbb{R}$ and $\delta \in (0, +\infty)$. We consider a continuous random variable X which is uniformly distributed on the interval $[T-\delta, T+\delta]$.

(i) (2 points) Give the probability density function of X.

(ii) (4 points) Compute the expectation $\mathbb{E}[X]$ of X.

(iii) (4 points) Compute the variance $\mathrm{Var}(X)$ of X.

(iv) (6 points) We now consider a sample of n i.i.d. random variables $(X_i)_{1 \leq i \leq n}$, each continuous with the same distribution as X, i.e. uniform on $[T-\delta, T+\delta]$. We want to estimate the two parameters $T \in \mathbb{R}$ and $\delta \in (0, +\infty)$, which are unknown. Find estimators of T and δ using the method of moments.

(v) (6 points) Compute the maximum-likelihood estimators $\hat{T}_{\mathrm{MLE}}$ and $\hat{\delta}_{\mathrm{MLE}}$, for T and δ respectively.

(vi) (8 points) We now let $n = 10000$, and we count the number N of observations, among $X_1, \ldots, X_n$, that fall in the subinterval $[T + \frac{\delta}{2}, T+\delta]$, i.e.

$$N = \left| \left\{ i \in \{1, \ldots, n\} \ : \ X_i \in \left[T + \frac{\delta}{2}, T+\delta \right] \right\} \right|.$$

Give an estimate of $\mathbb{P}(N \geq 2600)$ using the Central Limit Theorem, in terms of the cumulative distribution function Φ of a random variable with the standard normal distribution.